

The general Rankin–Selberg pair by polarization

Samuel Lavery*

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Abstract

For an arbitrary pair f, g of level-one cusp forms of one weight k , the completed Rankin–Selberg transform is constructed and its full analytic package proven — global self-dual functional equation, polar structure with residue the polarized Petersson pairing, entire part, Deligne-chart identification, and literal cross Hecke coefficients — by *polarization*: the cross object is $\frac{1}{4}[\Lambda_{f+g} - \Lambda_{f-g} + i\Lambda_{f+ig} - i\Lambda_{f-ig}]$, and the compiled single-form engine (the averaged lattice pair with its reflection from the theta transformation law) applies to each of the four diagonal forms verbatim. No new analytic input enters; everything is unconditional and machine-checked.

1 Polarization

Lemma 1 (sesquilinear polarization). *For $x, y \in \mathbb{C}$: $x\bar{y} = \frac{1}{4}[(\|x + y\|^2 - \|x - y\|^2) + i(\|x + iy\|^2 - \|x - iy\|^2)]$. (*polarization*.)*

Write $h_1 = f + g$, $h_2 = f - g$, $h_3 = f + ig$, $h_4 = f - ig$ — cusp forms of the same weight (`polForm1...4`). Since q -expansion is linear (the compiled expansion-linearity at level one), Lemma 1 applied coefficientwise gives $a_f(n)\overline{a_g(n)} = \frac{1}{4}[\|a_{h_1}(n)\|^2 - \|a_{h_2}(n)\|^2 + i\|a_{h_3}(n)\|^2 - i\|a_{h_4}(n)\|^2]$ (`cross_coeff_polarization`).

2 The cross pair

Let $\bar{\theta}_h$ be the compiled averaged lattice profile of a cusp form h and $m_h = \|h\|^2$ its Petersson mass. Define the *cross profile* and *cross mass*

$$\Theta_{f,g} = \frac{1}{4}[(\bar{\theta}_{h_1} - \bar{\theta}_{h_2}) + i(\bar{\theta}_{h_3} - \bar{\theta}_{h_4})], \quad m_{f,g} = \frac{1}{4}[(m_{h_1} - m_{h_2}) + i(m_{h_3} - m_{h_4})],$$

the second being the Petersson pairing $\langle f, g \rangle$ in polarized form. Each $\bar{\theta}_{h_j}$ satisfies the weight-one reflection $\bar{\theta}_{h_j}(1/t) = t\bar{\theta}_{h_j}(t)$ — the lattice theta transformation law carried through the Petersson average, form by form — so the combination satisfies it too, and each carries the compiled saddle decay and local integrability. Hence the data assemble into a Mathlib weak functional-equation pair with weight 1, root number 1, and both constant terms $m_{f,g}$ (`crossPair`), every field inherited from the four compiled diagonal pairs.

Theorem 2 (the analytic package). *Let Λ be the completed transform of the cross pair. Then, unconditionally: $\Lambda(1 - s) = \Lambda(s)$ for every $s \in \mathbb{C}$; $(s - 1)\Lambda(s) \rightarrow m_{f,g}$ as $s \rightarrow 1$; and Λ minus its two booked polar terms is entire. (`cross_selfdual_FE`, `cross_residue`, `cross_entire`.)*

*Machine-checked companion: `RequestProject/RankinSelbergCross.lean`, 13 named declarations, each at axiom footprint `{propext, Classical.choice, Quot.sound}` or a subset, no `sorry`, no `axiom`, zero `sorryAx`.

3 Chart and coefficients

Define the cross Rankin bank as the polarized combination of the four diagonal rank-4 banks, $c_{f,g} = \frac{1}{4}[(c_{h_1} - c_{h_2}) + i(c_{h_3} - c_{h_4})]$ (`crossRankinBank`).

Theorem 3 (chart identification). *On $2 < \Re s$: $\Lambda(s) = 2^{-k} \Gamma_{\mathbb{C}}(s) \Gamma_{\mathbb{C}}(s + k - 1) L(c_{f,g}, s)$.*

Proof. The pair's transform is the Mellin transform of $\Theta_{f,g} - m_{f,g}$; Mellin linearity (`mellin_add'`, `mellin_sub'`, with the four compiled convergences) splits it into the four diagonal tails; the compiled unpeeled landing evaluates each as $2^{-k} \Gamma_{\mathbb{C}}(s) \Gamma_{\mathbb{C}}(s + k - 1) L(c_{h_j}, s)$ — valid for *arbitrary* cusp forms, no eigenform input — and L-series linearity reassembles the combination. (`cross_mellin_split`, `LSeries_crossRankinBank`, `cross_lambda_eq`) $\square$

Theorem 4 (coefficient identification). $c_{f,g} = \mathbf{1}_{\square} \star b_{f,g}$, where $b_{f,g}(n) = a_f(n) \overline{a_g(n)} n^{-(k-1)}$ is the literal cross Hecke datum in Deligne normalization.

Proof. Each diagonal bank factors as $c_h = \mathbf{1}_{\square} \star b_h$ (Möbius inversion $\zeta \star \mu = \delta$ inside the compiled convolution presentation, `rankinBank_eq_sqIndicator_mul`); convolution is bilinear, so the combination is $\mathbf{1}_{\square} \star b_{f,g}$ with $b_{f,g}$ the polarized combination of the b_h — which is the stated pointwise datum by coefficient polarization. (`crossSquare`, `crossSquare_apply`, `crossRankinBank_eq_sq`) $\square$

Theorem 5 (rung 6, one theorem). *For every pair f, g of level-one cusp forms of weight $k \geq 0$: the functional equation, polar structure, entire part, chart identification, and coefficient identification of Theorems 2–4 hold simultaneously. (`rankin_selberg_rung`.)*

Remark 6 (register and falsifiers). *Proven and machine-checked: everything above, for arbitrary pairs — no eigenform hypothesis anywhere in this note; the diagonal $f = g$ recovers the symmetric-square companion's pair. Cited, not re-proved: that the object so constructed is the automorphic Rankin–Selberg convolution of the classical theory (Rankin, Selberg; Jacquet for $\mathrm{GL}(2) \times \mathrm{GL}(2)$) — the identification of coefficients, chart, functional equation and poles with the classical normalization is exactly what the theorems state. A counterexample would be a pair violating $\Lambda(1 - s) = \Lambda(s)$, or a level at which $c_{f,g}$ differs from $\mathbf{1}_{\square} \star (a_f \bar{a}_g n^{-(k-1)})$ — each formally stated, so a counterexample would require a kernel failure.*

References

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- [2] A. Selberg, *Bemerkungen über eine Dirichletsche Reihe, die mit der Theorie der Modulformen nahe verbunden ist*, Arch. Math. Naturvid. **43** (1940), 47–50.
- [3] H. Jacquet, *Automorphic forms on $\mathrm{GL}(2)$, Part II*, Lecture Notes in Math. **278**, Springer (1972).